All-Order Prime-Tail Integral Resummation

RH research program

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Abstract

Let $3 = p_1 < p_2 < \cdots$ be the odd primes, put $x = p_y$ and $L = \log x \geq 512$, and let

$$V = L^{3/5}(\log L)^{-1/5}, \qquad \varepsilon_x = 0.027L^{1.801}e^{-0.1853V}.$$

For $c = 1, \ldots, 7$ we resum the whole family of strict tails $P_r(y) = \sum_{p>x}(p^2-1)^{-r}$ into the coordinates $\Phi_c^P = \sum_{r\geq 1} c^r P_r/r$, and compare them with two integral coordinates Φ_c^J and Φ_c^I. Summing the absolute Stieltjes error before taking relative logarithms gives

$$\max_c |\Phi_c^P - \Phi_c^J| \leq \frac{28\varepsilon_x}{xL}, \qquad 0 \leq \max_c(\Phi_c^J - \Phi_c^I) \leq \frac{14}{3x^3L}.$$

The exact square-clock endpoint map is Lipschitz from ℓ^∞ to $\mathbb{R}$ with dual ℓ^1 gradient bound 126 on the required seven-dimensional cube. Consequently, for precisely defined gaps $\mathrm{Gap}_P, \mathrm{Gap}_J, \mathrm{Gap}_I$,

$$\pi^2|\,\mathrm{Gap}_P - \mathrm{Gap}_J\,| \leq \frac{3528\varepsilon_x}{xL}, \qquad \pi^2|\,\mathrm{Gap}_J - \mathrm{Gap}_I\,| \leq \frac{588}{x^3L}.$$

This is an infinite-order source–kernel exchange and endpoint Lipschitz theorem, not a termwise consequence of a finite-partition result. Since $\varepsilon_x x^2 \to \infty$, it gives no second-order or P_2-scale precision. A 42-row exact artifact reproduces the algebraic interfaces but is not an analytic proof.

Keywords: explicit prime number theorem; strict prime tails; all-order logarithmic resummation; endpoint map; uniform Lipschitz bound.

1 Statement and scope

Let $\vartheta(t) = \sum_{p\leq t}\log p$. Johnston and Yang prove the explicit Vinogradov–Korobov estimate used below in their Theorem 1.4, equation (1.8), printed page 2 [1]. In the present notation it yields

$$|\vartheta(t) - t| \leq t\varepsilon_t, \qquad \varepsilon_t = 0.027(\log t)^{1.801}\exp\Big[-0.1853(\log t)^{3/5}(\log\log t)^{-1/5}\Big] \tag{1}$$

for $t \geq 23$. Direct differentiation shows that ε_t decreases once $\log t \geq 512$; this is also the source transfer frozen in RH-386 [3]. Hence $|\vartheta(t) - t| \leq t\varepsilon_x$ for $t \geq x$.

For every integer $r \geq 1$ define

$$P_r(y) = \sum_{p>x}(p^2-1)^{-r}, \quad h_r(t) = \frac{1}{(t^2-1)^r\log t}, \quad J_r = \int_x^\infty h_r(t)\,dt, \quad I_{2r} = \int_x^\infty \frac{t^{-2r}}{\log t}\,dt. \tag{2}$$

The endpoint $p > x$ is strict throughout. For $1 \leq c \leq 7$ set

$$\Phi_c^P = \sum_{r\geq 1}\frac{c^r P_r(y)}{r}, \tag{P}$$

$$\Phi_c^J = \int_x^\infty \frac{-\log(1 - c/(t^2-1))}{\log t}\,dt, \tag{J}$$

$$\Phi_c^I = \int_x^\infty \frac{-\log(1 - c/t^2)}{\log t}\,dt. \tag{I}$$

We use the corresponding undecorated symbols for the seven-vectors:

$$\Phi^P = (\Phi_1^P, \dots, \Phi_7^P), \qquad \Phi^J = (\Phi_1^J, \dots, \Phi_7^J), \qquad \Phi^I = (\Phi_1^I, \dots, \Phi_7^I). \tag{3}$$

The exact endpoint map F and the three gaps are defined in Section 5. The main result is the following.

Theorem 1.1 (All-order integral resummation)**.** *For every y with $x = p_y$ and $L = \log x \geq 512$, all three coordinate vectors lie in $[0, 1/2]^7$, and*

$$\max_{1\leq c\leq 7} |\Phi_c^P - \Phi_c^J| \leq \frac{28\varepsilon_x}{xL}, \tag{4}$$

$$0 \leq \max_{1\leq c\leq 7} (\Phi_c^J - \Phi_c^I) \leq \frac{14}{3x^3L}. \tag{5}$$

Moreover

$$\pi^2 |\operatorname{Gap}_P - \operatorname{Gap}_J| \leq \frac{3528\varepsilon_x}{xL}, \tag{6}$$

$$\pi^2 |\operatorname{Gap}_J - \operatorname{Gap}_I| \leq \frac{588}{x^3L}, \tag{7}$$

$$\pi^2 |\operatorname{Gap}_P - \operatorname{Gap}_I| \leq \frac{3528\varepsilon_x}{xL} + \frac{588}{x^3L}. \tag{8}$$

All constants are uniform in the seven real integer channels $c \in \{1, \dots, 7\}$.

2 Strict Stieltjes transfer before resummation

Write $E(t) = \vartheta(t) - t$. Stieltjes integration at the strict endpoint gives

$$P_r = \int_{(x,\infty)} h_r(t)\, d\vartheta(t) = -\vartheta(x)h_r(x) - \int_x^\infty \vartheta(t)h_r'(t)\, dt, \tag{9}$$

$$J_r = -xh_r(x) - \int_x^\infty t h_r'(t)\, dt. \tag{10}$$

Thus no prime at x is included and

$$P_r - J_r = -E(x)h_r(x) - \int_x^\infty E(t)h_r'(t)\, dt. \tag{11}$$

Since h_r decreases, (1) and (10) imply the absolute bound

$$|P_r - J_r| \leq \varepsilon_x\{2xh_r(x) + J_r\}. \tag{12}$$

This is the crucial order of operations: we will sum (12) over every r directly. The relative estimate $|\log(P_r/J_r)| \leq 14r\varepsilon_x$ from RH-386 has an r-dependent smallness condition and cannot be summed over all orders.

Proposition 2.1 (All-order source coordinate)**.** *For $1 \leq c \leq 7$,*

$$|\Phi_c^P - \Phi_c^J| \leq \frac{3c\varepsilon_x}{xL\{1 - (1+c)/x^2\}} < \frac{4c\varepsilon_x}{xL}. \tag{13}$$

Proof. All summands are nonnegative and $c/(p^2-1) \leq 7/24 < 1$. Tonelli's theorem and $-\log(1-z) = \sum_{r\geq 1} z^r/r$ give

$$\Phi_c^P = \sum_{p>x} -\log\left(1 - \frac{c}{p^2-1}\right), \qquad \Phi_c^J = \sum_{r\geq 1} \frac{c^r J_r}{r}, \tag{14}$$

and the latter sum is precisely the integral in (J). Now sum (12) with weights c^r/r. At the boundary,

$$\sum_{r\geq 1}\frac{c^r h_r(x)}{r}=\frac{-\log(1-c/(x^2-1))}{L}\leq\frac{c}{L(x^2-1-c)}.$$

In the integral, $-\log(1-z)\leq z/(1-z)$ and $t\geq x$ give

$$\Phi_c^J\leq\frac{c}{L\{1-(1+c)/x^2\}}\int_x^\infty t^{-2}\,dt=\frac{c}{xL\{1-(1+c)/x^2\}}.$$

The two boundary units and one integral unit total the first constant in (13). Since $x>256$ and $c\leq 7$, the denominator is greater than $3/4$, proving the strict $4c$ bound. Taking $c\leq 7$ yields (4). ☐

3 Power-kernel resummation

The same nonnegative Tonelli argument gives the second integral identity

$$\Phi_c^I=\sum_{r\geq 1}\frac{c^r I_{2r}}{r}. \tag{15}$$

Proposition 3.1 (Exact-to-power coordinate)**.** *For* $1\leq c\leq 7$,

$$0\leq\Phi_c^J-\Phi_c^I\leq\frac{c}{3x^3L\{1-(1+c)/x^2\}}<\frac{2c}{3x^3L}. \tag{16}$$

Proof. The difference of the two logarithmic integrands is nonnegative, and

$$\begin{aligned}\log\frac{1-c/t^2}{1-c/(t^2-1)}&=\log\left(1+\frac{c}{t^2(t^2-1-c)}\right)\\&\leq\frac{c}{t^2(t^2-1-c)}\leq\frac{ct^{-4}}{1-(1+c)/x^2}.\end{aligned}$$

Use $1/\log t\leq 1/L$ and $\int_x^\infty t^{-4}dt=1/(3x^3)$. Because $x>256$ the displayed denominator exceeds $1/2$. Maximizing $2c/3$ over $c\leq 7$ proves (5). ☐

4 The real coordinate cube

The analytic domain is deliberately much smaller than the source theorem's $x\geq 23$. Since $L\geq 512$,

$$x=e^L>2^L\geq 2^{512}>256. \tag{17}$$

For integral x the telescoping identity

$$\sum_{n>x}\frac{1}{n^2-1}=\frac{1}{2}\left(\frac{1}{x}+\frac{1}{x+1}\right) \tag{18}$$

and $-\log(1-z)\leq 2z$ for $0\leq z\leq 1/2$ show

$$0\leq\Phi_c^P\leq c\left(\frac{1}{x}+\frac{1}{x+1}\right)<\frac{2c}{x}<\frac{14}{256}<\frac{1}{2}.$$

The direct integral bounds used above give the same conclusion for Φ_c^J and Φ_c^I (also $0\leq\Phi_c^I\leq\Phi_c^J$). Therefore all three vectors and every line segment joining two of them belong to the convex cube $[0,1/2]^7$.

5 Exact endpoint map and its Lipschitz ledger

For $2 \le m \le 8$ put

$$u_m = \prod_{\substack{p \text{ odd} \\ p \text{ prime}}} \frac{1 - m/p^2}{1 - 1/p^2} = \prod_{\substack{p \text{ odd} \\ p \text{ prime}}} \left(1 - \frac{m-1}{p^2 - 1}\right). \tag{19}$$

For fixed $m \le 8$,

$$\sum_{p \text{ odd}} \frac{m-1}{p^2-1} < \infty.$$

Thus the standard infinite-product criterion shows that (19) converges to a positive value, not merely a product of positive finite factors. All factors after $p = 3$ are at most one, so retaining the $p = 3$ factor gives

$$0 < u_m \le \frac{9-m}{8}. \tag{20}$$

Index the endpoint arrays by $m = 2, \dots, 8$:

$$(\alpha_m) = (-2, 2, -2, 2, -2, 2, -2), \qquad (\beta_m) = (1, -2, 2, -2, 2, -2, 2), \tag{21}$$

and define

$$C(V) = 1 + \sum_{m=2}^{8} \alpha_m V_m, \qquad W(V) = \sum_{m=2}^{8} \beta_m V_m.$$

For $z = (z_1, \dots, z_7) \in \mathbb{R}^7$, write $V_m(z) = u_m e^{z_{m-1}}$ and set

$$F(z) = 2\{C(u) - C(V(z))\} - 4W(V(z))(1 - e^{-z_1}). \tag{22}$$

The exact endpoint normal form of RH-383 [2] applies to (P). With

$$q_y = 4 \prod_{i \le y} p_i^2, \qquad \mathrm{Gap}_P := B_\infty - G(q_y) = \frac{F(\Phi^P)}{\pi^2}, \qquad \mathrm{Gap}_J := \frac{F(\Phi^J)}{\pi^2}, \qquad \mathrm{Gap}_I := \frac{F(\Phi^I)}{\pi^2}, \tag{23}$$

Gap_P is the actual square-clock gap, while Gap_J and Gap_I are the two integral surrogates introduced here. These are the only three gap objects used in the theorem.

Lemma 5.1 (Endpoint Lipschitz bound)**.** *For every $z \in [0, 1/2]^7$,*

$$\|\nabla F(z)\|_1 < 126. \tag{24}$$

Consequently $|F(z) - F(z')| \le 126\|z - z'\|_\infty$ for any two points of the cube.

Proof. The bound (20) and the exact arrays give the two ledgers

$$A := \sum_{m=2}^{8} |\alpha_m| u_m \le 7, \qquad B := \sum_{m=2}^{8} |\beta_m| u_m \le \frac{49}{8}. \tag{25}$$

Differentiate (22). The C term, the derivative falling on W, and the derivative falling on $1 - e^{-z_1}$ have respective coefficients

$$2, \qquad 4, \qquad 4. \tag{26}$$

On the cube, $V_m(z) \le e^{1/2} u_m$, $0 \le 1 - e^{-z_1} \le 1$, and $e^{-z_1} \le 1$. Therefore the three contributions to the ℓ^1 gradient are bounded by $2e^{1/2}A$, $4e^{1/2}B$, and $4e^{1/2}B$. Since $e^{1/2} < 2$,

$$\|\nabla F(z)\|_1 \le e^{1/2}\left(2 \cdot 7 + 4 \cdot \frac{49}{8} + 4 \cdot \frac{49}{8}\right) = 63e^{1/2} < 126.$$

The final assertion is the mean-value theorem along the segment from z to z', with the ℓ^∞ input norm paired with its dual ℓ^1 gradient norm. $\square$

6 Proof of the main theorem

The coordinate assertions and the cube membership were proved in Propositions 2.1 and 3.1 and Section 4. Apply Lemma 5.1 on the segment from Φ^P to Φ^J. Equations (23) and (4) give

$$\pi^2|\operatorname{Gap}_P - \operatorname{Gap}_J| = |F(\Phi^P) - F(\Phi^J)| \le 126\,\frac{28\varepsilon_x}{xL} = \frac{3528\varepsilon_x}{xL}.$$

The same argument between Φ^J and Φ^I gives

$$\pi^2|\operatorname{Gap}_J - \operatorname{Gap}_I| \le 126\,\frac{14}{3x^3L} = \frac{588}{x^3L}.$$

Their sum proves (8) and completes the proof of Theorem 1.1.

7 Novelty and sharp scope

The preceding proof is not obtained by inserting every r into the growing-order partition theorem of RH-386. That theorem controls a finite partition through a relative logarithmic estimate whose source condition includes $7R\varepsilon_x \le 1/2$, where R is the largest active order. The vector (P) contains every $r \ge 1$, so no finite R exists. Here the absolute strict Stieltjes error (12) is summed first by Tonelli; only then is the result passed through the entire, partition-infinite endpoint map by the new uniform Lipschitz bound. This simultaneous r-infinite and partition-infinite source–kernel exchange is the genuine new theorem.

The available source error does not resolve the next power-kernel scale. Indeed

$$\log(\varepsilon_x x^2) = 2L + \log(0.027) + 1.801\log L - 0.1853V(L) \longrightarrow +\infty, \tag{27}$$

because $V(L) = o(L)$. Thus the source term $\varepsilon_x/(xL)$ is asymptotically larger than the $r = 2$ scale $1/(x^3L)$. In particular, Theorem 1.1 claims neither a second-order coefficient nor P_2-scale (and hence no cubic) precision.

All channels in this paper are the seven real integers $1 \le c \le 7$. There is no complex-c extension, no prefix/prime-index simultaneous limit exchange, and no clock varying with a prefix length. The frozen phasewise class retains $c_{11} = 0$; active c_{11} is outside the source theory. The result supplies no adaptive capacity and no operator, trace, or prime-power trace; it makes no zeta-zero identification and does not prove the Riemann hypothesis. Gates A–E are all false.

8 Executable reproduction and source integrity

The companion exact artifact has epistemic role `reproduction_not_analytic_proof`. Its 42 oracle rows comprise 12 analytic/source-interface rows, seven channel rows, seven endpoint rows, 14 finite formal-resummation rows, and two ledger rows. They independently recompute the exact fractions, the cube bridge, the $2, 4, 4$ derivative ledger, 126, 3528, and 588. The 14 finite rows are degree-four truncations of the formal logarithm/product compiler, but they neither truncate nor prove the analytic theorem.

Twenty-four genuine field-level mutations are rejected, including altered Vinogradov–Korobov constants, a reversed log–log exponent, the wrong domain or channel range, an inclusive prime endpoint, a missing series divisor or Stieltjes boundary, false $3c$ and 21 source constants, interchanged J/I kernels, wrong endpoint signs, deletion of the final loss derivative, the wrong norm pair, and an illicit claim of P_2 precision. Strict JSON parsing rejects duplicates, nonfinite values, wrong top-level types, and Boolean-for-integer aliases. These tests check the algebraic declaration; Johnston–Yang's estimate, Stieltjes integration, Tonelli's theorem, and the mean-value argument remain the analytic proof above.

The immutable evidentiary closure contains 68 Git blobs: 59 inherited RH-386 source rows, the eight standard RH-386 release files, and its external-lock blob. One canonical remote Johnston–Yang lock makes 69 logical sources. The author-manuscript PDF and source tar are not vendored or archived. Network verification is disabled by default and is available only as an explicit opt-in check of the versioned URLs, redirect policy, PDF MIME type, byte counts, page count, and SHA-256 values.

The immutable source identifiers are, in order, the RH-386 release commit, the ordered 68-Git-row digest, the 69-logical-source digest, and the canonical remote-lock digest:

```
9778e3515d45816665d672a641947b93906abf54
19def5cbed919da8e9652012cf011f3b5728efd4b24a9eef0911bb7346467d27
5016397fe59962954514b3b42d68e9de6dfeff0dae949791b01c6a516f5c61fe
d53b93212b7c5b5b6b3f7e890099c48ce8e35f2bff9bdd49f9c330a9b5039786
```

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Declarations

Data and code availability. The proof, exact certificate compiler, strict tests, official closed Draft 2020–12 schema, source-lock records, offline-by-default verifier, and release replay tools accompany the paper. The Johnston–Yang PDF and source tar are not redistributed; their exact remote identifiers are recorded in the external lock.

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AI assistance. AI-assisted tools were used for symbolic-interface enumeration, adversarial testing, manuscript drafting, and release auditing. All mathematical claims, citations, source locks, and final text were reviewed under author responsibility.

References

[1] Daniel R. Johnston and Andrew Yang. Some explicit estimates for the error term in the prime number theorem. *Journal of Mathematical Analysis and Applications*, 527(2):127460, 2023. doi: 10.1016/j.jmaa.2023.127460.

[2] RH research program. An exact euler-tail partition normal form for the square-clock gap, 2026. Frozen repository release RH-383.

[3] RH research program. Vinogradov–korobov growing-order prime-tail uniformization, 2026. Frozen repository release RH-386.